

# IN6227 Data Mining - Assignment 1

Variant-2

IN6227-Assignment-1

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Model name & version: deepseek/deepseek-flash LLM interface: Agent Harness such as opencode 1.18.34

Dataset: IN6227 A1 Variant 2 - mice\_protein.csv (1,080 rows; 72 mice, 8 classes)

## 1. Data exploration and cleaning

**Dataset.** mice\_protein.csv - 1,080 rows, 82 feature columns. The label is class with 8 classes, distributed c-CS-m 150 / c-SC-m 150 / c-CS-s 135 / c-SC-s 135 / t-CS-m 135 / t-SC-m 135 / t-SC-s 135 / t-CS-s 105 - a majority:minority ratio of 1.43:1, which is what the choice of metric in section 4 turns on. 77 numeric and 0 categorical columns were modelled; 5 further column(s) were held out of the feature set: MouseID, mouse\_id, Genotype, Treatment, Behavior.

**Issues found.** Missing values: 1,396 cells across 49 column(s), worst BCL2\_N (285); duplicate rows: none; outliers: kept - no row was filtered, so no fence was applied; class imbalance: 1.43:1.

### Cleaning decisions.

| Issue                                                          | Action                                                  | Why                                                                                                  |
|----------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Genotype/Treatment/Behavior define class; mouse_id repeats 15x | all held out of features; mouse_id used as group_column | class is their cross-product (a screen with them scored 1.000) and an entity key is not a predictor. |
| 1,396 missing cells; 7-mouse rarest class; heavy tails         | in-pipeline imputation; class and outliers kept         | imputation fitted per fold; the class is real (lift 10.4) and extreme ratios can be signal.          |

**After cleaning.** 1,080 rows loaded; none removed; 1,080 rows modelled, 77 features.

## 2. Feature selection / engineering

No features were built: the 77 columns are already normalised protein ratios, and ratios or interactions would multiply a 77-wide, 72-entity design that the RBF kernel and L2 penalty already partly absorb. The model sees 77 numeric features.

## 3. Model training

**Candidate-model reasoning.** The task is to recover the eight-way class from 77 protein measurements. The frame's Genotype, Treatment and Behavior define class exactly; an initial row-level screen including them scored f1\_macro 1.000 - leakage, not skill. Holding those out and grouping on mouse\_id, the seven families screen entity-disjointly as logistic\_regression 0.543, svm 0.528, random\_forest 0.472, gradient\_boosting 0.416, naive\_bayes 0.392, decision\_tree 0.376, knn 0.371. The one-SD shortlist is logistic\_regression + svm - the only pair in it with different inductive biases (linear log-odds against an RBF margin), so that pair is taken.

|                       | Model A                                                                                                                                             | Model B                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Estimator             | Logistic regression (LogisticRegression via logistic_regression)                                                                                    | Support vector machine (SVC via svm)                                                                                                                |
| Hyperparameters       | C=10.0, class_weight=None - best of 6 combination(s) scored                                                                                         | C=10.0, gamma=0.001 - best of 9 combination(s) scored                                                                                               |
| Tuned on              | 3-fold cross-validation over the 855 pool rows, scored on f1_macro                                                                                  | 3-fold cross-validation over the 855 pool rows, scored on f1_macro                                                                                  |
| Validation score      | 0.5592 (f1_macro)                                                                                                                                   | 0.5738 (f1_macro)                                                                                                                                   |
| Stopping criterion    | lbfgs, max_iter=2000, tol=1e-4; no early stop                                                                                                       | libsvm QP to tol=1e-3, max_iter=-1                                                                                                                  |
| Overfitting control   | L2 penalty; C tuned 0.1-10                                                                                                                          | maximal margin, RBF; C and gamma tuned                                                                                                              |
| Preprocessing applied | numeric: median imputation, then standardised; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros) | numeric: median imputation, then standardised; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros) |

### Preprocessing, per model.

- logistic\_regression: generally recommended, close to required in practice - a shared L2 penalty compares coefficients, so units matter.
- svm: generally recommended, close to required for an RBF kernel - the margin is a distance.

**Fixed configuration (not tuned):** seed 42; test fraction 20% of the frame; validation by 3-fold cross-validation over the pool; tuning metric f1\_macro; primary metric f1\_macro.

## 4. Evaluation and comparison

Split protocol: 855 pool rows, validated by 3-fold cross-validation, plus 225 test rows. The test set is held out by whole entity, so no entity is on both sides (20% of mice\_protein.csv), seed 42, and was not touched again until the final scoring.

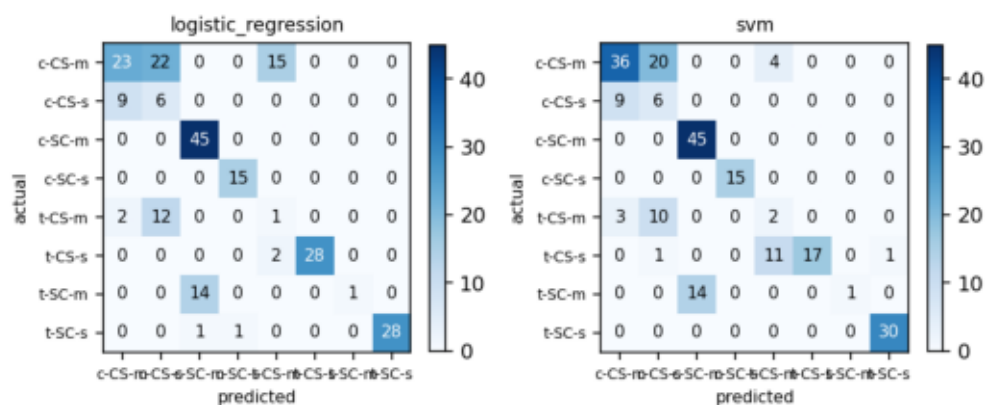
Validation strategy: stratified, entity-disjoint k-fold cross-validation, splitting at the entity unit. Chosen because The split unit is the mouse: 1080 rows are 72 mice x 15, and sharing a mouse\_id makes rows near-identical (nearest-neighbour share 0.994, lift 76.6), so a row split would reward recognition - the first screen's perfect scores. stratified\_group\_kfold on mouse\_id keeps whole mice on one side with class proportions kept. The rarest class t-CS-s has 7 mice; after the 20% grouped test holdout 5 remain, so each of three folds validates one mouse (15 rows) of it, all eight classes appear per fold, and no mouse is shared. Validation class counts: c-CS-m 30 / c-CS-s 40 / c-SC-m 35 / c-SC-s 40 / t-CS-m 40 / t-CS-s 25 / t-SC-m 40 / t-SC-s 35 (mean per fold, over 3 folds - a mean rather than a count, so it need not be a whole number). Cost of the choice: 3x fitting; every row is validated against exactly once, so the score is the mean of 3 estimates rather than a single one, and the final model still trains on all 855 rows.

Validation used for: hyperparameter selection. Final model fitted on: the whole pool (855 rows) - with 3-fold cross-validation every row is validated against exactly once, so final\_fit\_scope: pool fits the same rows as train would, and the test set is the only estimate untouched by selection.

| Metric               | Model A      | Model B      |
|----------------------|--------------|--------------|
| Accuracy             | 0.6533       | 0.6756       |
| Precision (macro)    | 0.6962       | 0.7200       |
| Recall (macro)       | 0.5979       | 0.5958       |
| F1 (macro) (primary) | 0.5811       | 0.5900       |
| ROC-AUC / PR-AUC     | 0.9317 / n/a | 0.9410 / n/a |

AUC notes: not defined for multiclass (expected, not a failure).

Confusion matrices:



**Primary metric and rationale.** f1\_macro. f1\_macro: assign samples to eight groups with none the special positive and equal cost for adjacent-group swaps, so classes should weigh equally. The classes are 105-150 rows (ratio 1.43), so accuracy would let a model ignore t-CS-s, and all eight appear in every fold. accuracy was not led with: Accuracy is majority-weighted; a model that never predicts t-CS-s could still score near 0.9, hiding exactly the failure a classifier of a rare experimental group must not hide.

## 5. Findings and discussion

**Performance comparison.** The SVM leads on every headline metric: f1\_macro 0.590 vs 0.581, accuracy 0.676 vs 0.653, macro ROC-AUC 0.941 vs 0.932. The linear model is steadier fold-to-fold (screen SD 0.051 vs 0.073) but confuses c-CS-m with c-CS-s; the RBF boundary separates the c-SC/t-SC groups exactly and lifts c-CS-m recall to 0.60 at the cost of more t-CS-s misroutes.

**Limitations.** The estimate rests on 72 mice, not 1,080 independent rows: the rarest class is 7 mice, so one mouse crossing a fold moves its recall by up to one seventh, and the ~14-mouse test holdout may rest a class on one animal. Correlated, heavy-tailed protein columns make the linearity and distance assumptions approximate, so the comparison is descriptive rather than causal.

## Repository

<https://github.com/NIQIXINNN/IN6227-assignment>